

Capitalization

Titles

- Sentence case (as of 5/23/14 issue), but cap the first word after a colon
- Do not cap articles “a,” “an,” or “the,” except after a colon or at the beginning of a title.
- Lowercase *cis*-, *trans*-, *tert*-, *sec*-, and similar prefixes.
- Cap genus but lowercase species when they appear together in a title.

Ex: Competition between *Aedes albopictus* and *Aedes triseriatus*

Subheads

In Reviews, Research Articles, and Perspectives,

- for first-level subheads, use sentence case and boldface on a separate line.
- for second-level subheads, use sentence case, boldface, and italics, also on a separate line.
- for third-level subheads, use sentence case and italics, also on a separate line.

Persons

- Cap titles that appear immediately before the name (as part of the name) but not after a name.

Ex: Secretary of State Warren Christopher; Christopher, the secretary
of state
Senator Kennedy; the senator from Massachusetts
President Clinton; Clinton, president of the United States
former president George Bush
the president of Harvard
Frank Press, president, National Academy of Sciences
Science Adviser John Gibbons
John Gibbons, the presidential science adviser

Exception: If the title before a name is used in apposition to the name, do not cap it.

Ex: The senator from Massachusetts, John F. Kennedy, said that
The secretary of state, W. Christopher, is alleged
EPA's deputy administrator, A. James Barnes, said the agency
NIH Director Healy said that
(Also, see *Chicago Manual of Style*, 14th Edition, 7.16 to 7.24)

- Do not cap unofficial titles.

Ex: astronaut Smith

- In names of peoples and races:

Cap: Aboriginal, Black, Caucasian, Hispanic, Latino/a/x, Indian,
Indigenous
Note that Hispanic and Latino/a/x are not necessarily interchangeable.

Lowercase: white

Government, Titles, and Offices

the administration; the Clinton administration
the Cabinet, Cabinet-level position
the Constitution, constitutional
Cold War
Executive Branch, Executive Budget
the president (United States), presidential
federal government, US government; British government
the Senate, the senators
Congress, congressional, congressman, the house of Congress
committee, subcommittee (cap when exact name is used)
Nobel Prize (when referring to particular prize), Nobel laureate
the internet
the act (Wild Free-Roaming Horses and Burros Act)

Organizations, Institutions, and Companies

- Cap complete title of organization; use lowercase when only part of the title is used to refer to the organization.

Ex: Centers for Disease Control, the centers
National Institutes of Health, the institutes
* National Academy of Sciences, the academy

- Full titles of institutions and companies and the names of their departments and divisions are capitalized, but generic or descriptive terms, such as “mathematics department,” are lowercased when used alone.

Ex: the University of Chicago, the Department of History
the university, the history department

The word “the” at the beginning of such titles is capitalized only when the official corporate name of the institution is called for.

Ex: © 1982 by The University of Chicago

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But: John Martin is with the Department of Oncology at the University of Chicago.

Geography and Directions

- In general, cap north, south, northeast, northern, etc., when they designate regions (that is, the region can be located on a map), but use lowercase when they indicate compass directions.

Ex: the South (U.S.), southern France, south of the city, a Southerner (cap only if used in Civil War context), Southeast Asia, south-southeast.

Northern Hemisphere, North Atlantic, but west Atlantic.

the West, Western countries (political division), western Ohio, Western Europe.

East Coast (of United States), on the Atlantic Coast (if referring to region, otherwise lowercase), eastern Atlantic.

the Midwest (U.S.), midwestern town, a midwesterner.

Central America, central Asia, central Europe (Central Europe, if referring to World War I political division).

the Arctic, arctic weather.

Cities and States

* Use New York state (lowercase state) when it is not clear from context whether reference is to city or state.

But: New York City, city of New York, state of New York.

- Cap “the” when it is part of a name.

Ex: The Hague

But: the Bronx, the Netherlands [but just “Netherlands” (no “the”) in addresses]

Rivers and Other Generic Terms

Use the following style:

Amazon River, Rhine River valley
the river Danube (“the” precedes generic term)
the Missouri and Mississippi rivers (lowercase term with plurals
after the proper name)
Lakes Geneva and Lucerne (cap term preceding plurals)
Gale crater
Kilauea Iki crater, Nevada del Ruiz volcano
Mississippi River delta
Missouri River basin
San Andreas fault
Greenland Ice Sheet

Terms Involving Proper Nouns

- Cap only proper nouns in names of laws and theories.

Ex: Planck’s constant
Boyle’s law
Avogadro’s number

- Adjectives derived from proper nouns may be cap or lowercase, depending on whether they have a “proper” meaning or a “common” meaning.

Ex: Gaussian curve
Roman civilization
roman type
Wagnerian genetics
Cartesian coordinates

(In general, use *Webster’s III* if there is a question about capping terms involving proper nouns. See also: *Words into Type*, p. 148; *Chicago*, 7.49, 7.117 and, 7.120; and *ACS*, p. 86.)

Scientific Terms in Text

Medical

- Cap proper noun in names of diseases, but lowercase terms such as “disease” and “syndrome”; lowercase name of disease when used as an adjective.

Ex: Parkinson’s disease
 parkinsonian
 Down syndrome
 Wilms tumor

(For list of examples, see *Words into Type*, p. 149, and *Chicago*, 7.117; also *Dorland’s Medical Dictionary* has an extensive list.)

- Lowercase stereotaxic measurements (for instance, in the brain)

Ex: bregma (See *Stedman’s Medical Dictionary*)

Science style: x-ray
 petri dish

Gene and Protein names

- Capitalization varies depending on organism.
- Viral genes are lowercase and italic, but their proteins are in roman with an initial cap; regions of a virus (such as LTRs) are roman

Plants:

Arabidopsis thaliana:

-Genes: symbols consist of three (occasionally two) italicized uppercase letters, e.g., *EMB*, *DET*. Different loci with the same three-letter symbol are distinguished by different numbers, e.g., *EMB1*, *EMB2*, etc.

-Proteins: designation in capital letters and not italicized, e.g., DET1.

Yeast:

S. cerevisiae:

-Genes: three italic letters followed by an italicized arabic numeral, e.g., *ARG2*, *arg2 ade5*, *cdc28*. On the genetic map, the locus is named after the mutant phenotype, and because most mutations studied are recessive, almost all locus designations are lowercase, e.g., *arg2*.

-Proteins: same characters as gene symbol in roman type, with initial capital letter (Arg); superscript plus or minus indicates the independence of, or requirement for, a substance, respectively, e.g., Arg⁺ [does not require arginine]

C. elegans

Genes: 3 lowercase italic letters, a hyphen, and an arabic no., e.g., *dpy-5*, *let-37*

Proteins: the relevant gene name in nonitalic capital letters, e.g., the protein encoded by *unc-13* is UNC-13

Drosophila:

genes:

- symbols vary depending on whether allele is recessive to wild type (begins with lowercase letter, e.g., *awd* for *abnormal wing disc*) or dominant to wild type (begins with uppercase letter, e.g., *R* for *Roughened*)
- Genes named after a protein begin with an uppercase letter, eg. *Adh* (for alcohol dehydrogenase)
- different loci with similar function: gene symbol plus italic arabic numerals and capital letters; hyphens not used (e.g. *Act5C*, *Act42A*)
- Proteins: same as gene symbol but in nonitalic capital letters, e.g. HH (protein encoded by *hh* gene)

Mouse

Genes: 2, 3 or 4 italic letters and numbers, e.g. *Re*, *G6pd*, *B2m*; the first letter is uppercase.

Proteins: same symbol as the gene, all-uppercase, no italics; e.g., the product of the *Myc* gene is MYC.

Human:

Genes: uppercase italic letters or combination of uppercase letters + arabic numerals (*ADH*, *ACOI*); Greek letters, roman numerals, superscripts, subscripts, and punctuation are not acceptable

Proteins: same characters used for the loci encoding them, in nonitalic type

Here are online resources for more specific genetic and protein guidelines.

The Arabidopsis Information Resource Nomenclature Guidelines

<http://www.arabidopsis.org/portals/nomenclature/guidelines.jsp>

UniProtKB/Swiss-Prot nomenclature <http://www.expasy.ch/cgi-bin/lists?nomlist.txt>

Human Gene Nomenclature Committee <http://www.genenames.org/guidelines.html>

Human Genome Variation Society <http://www.hgvs.org/mutnomen/recs.html>

Genetic Nomenclature for *Drosophila melanogaster*

<http://flybase.org/wiki/FlyBase:Nomenclature>

International Union of Pure Applied Chemistry

<http://old.iupac.org/publications/epub/index.html#nt>

RCSB Protein Data Bank <http://www.rcsb.org/pdb/home/home.do>

Mouse Genome Informatics <http://www.informatics.jax.org/mgihome/nomen/gene.shtml>

Candida Genome Database <http://www.candidagenome.org/Nomenclature.shtml>

Saccharomyces Genome Database <http://www.yeastgenome.org/>

Genetic (and organism) nomenclature for bacteria http://jb.asm.org/site/misc/journal-ita_nom.xhtml

Chemical and Drug Names

- Lowercase chemical elements and compounds.

Ex: sodium chloride

- Nonproprietary drug names are not capped. If such a word is the first word in a sentence or in a title, use initial cap for the drug but do not cap the locant or positional prefix.

(See *ACS Style Guide*, p. 86)

- In sentences beginning with a symbol or element, cap the first word that follows.

Ex: *tert*-Butyl fluoride is
D-Ascorbic acid was
o-Dichlorobenzene is

(For a listing of capitalization styles for chemical names that appear at the beginning of a sentence or in a title, see *ACS Style Guide*, pp. 87–90.)

- Amino acids should be lowercase when spelled out (L-proline) but capped when abbreviated (L-Pro)
- One-letter: no superscripts (e.g., L323)
- Three-letter: with superscripts (e.g., Leu³²³)
- I307C means substitution of Ile³⁰⁷ by Cys; can define as Ile³⁰⁷→Cys
- N176^{5.39} (Ballesteros-Weinstein notation for G protein-coupled receptor residues)

A	Ala		alanine
C	Cys		cysteine
D	Asp		aspartic acid
E	Glu		glutamic acid
F	Phe		phenylalanine
G	Gly		glycine
H	His		histidine
I	Ile		isoleucine
K	Lys		lysine
L	Leu		leucine
M	Met		methionine
N	Asn		asparagine
P	Pro		proline
Q	Gln		glutamine
R	Arg		arginine
S	Ser		serine
T	Thr		threonine
V	Val		valine
W	Trp		tryptophan
X			unknown/unspecified
Y	Tyr		tyrosine

Plants and Animals

- Cap genus (italic); lowercase species (italic).

Ex: *Saccharomyces cerevisiae*

- Cap the names of all divisions higher than species—that is, kingdom, phylum, class, order, family (roman), and genus; lowercase adjectives and English nouns derived from them:

Chiroptera, chiropter

Coelenterata, coelenterate

Protozoa, protozoan

the genus *Hydra*; a hydra

the tall buttercup (*Ranunculus acris*)

the harp seal (*Phoca graenlandica*)

Parasitism is emphasized in the study of Sporozoa, Platyhelminthes, and Nematelminthes.

Such terms as “Metazoa” and “Protozoa” are properly capped as divisions of the animal kingdom; however, the plurals of the lowercase common-noun designations “metazoan” and “protozoan” are “metazoa” and “protozoa,” so that when used with the article “the” (that is, when speaking of members of the division and not the division itself), the proper style is the “metazoa” and the “protozoa.” Also, sometimes confusing is “Primates,” capped as the name of an order, and “primate” or “primates,” lowercase as members of the order.

(from *Words into Type*, p. 169)

Coronavirus variants

For SARS-CoV-2 variants, alphanumeric lineage designations (e.g., B.1.1.7) should always be used; Greek letter designations (e.g., Alpha) may also be used. However, the Greek letter designations should not replace the existing scientific names (for details and lists of variants, see [here](#) and [here](#)).

When the Greek letter designations are used, they should be capitalized, spelled out, and enclosed in parentheses after the lineage designation—e.g., “B.1.1.7 (Alpha).” Avoid using the country of presumed origin (use Greek letter designation instead).

Astronomy

-Cap “Earth” when used in any way except to mean “dirt” and use just “Earth,” not “the Earth.”

Ex: Studies about Venus, Mars, Earth
Researchers studied Earth’s mantle.
Studies of Earth’s atmosphere

But: He turned over the earth with his plow.

-Cap when referring to our own: Sun, Moon, Milky Way, Galaxy, Solar System, Universe, Galactic Center

-Cap: Big Bang, Kuiper Belt

-Lowercase when not referring to ours: sun, moon, galaxy, solar system (although other such systems are more likely to be identified as “planetary systems”), and universe (e.g., a simulated universe with alternative cosmological parameters)

-Comets: lowercase when “comet” precedes the name (e.g, comet Bennett, comet Halley), but cap when it follows (e.g., Halley’s Comet)

-Lowercase adjectival forms of planet names: martian, venusian, etc.

-Terms such as “crater” and “basin”: Capitalization depends on the official name of the feature. (In “Gale crater,” the word “crater” is lowercase because the formal name is “Gale,” not “Gale crater.” However, the mountain in the middle of that crater is officially named “Aeolis Mons,” so “Mons” is capitalized.) Names should be verified with the IAU Gazetteer of Planetary Nomenclature (<https://planetarynames.wr.usgs.gov/>).

-Classes of exoplanets (e.g., “hot jupiters,” “mini-Neptunes,” “super-Earths”): No standard convention for these (yet), so follow the author.

-Use parentheses around the numerical portion of asteroid names
e.g., “(486958) Arrokoth” and “(162173) Ryugu”

From Linda Rowan's "Celestial Bodies" style sheet (I:/lrowan/scistyle/cbodies.style):

"Capitalization:

Cap Sun, Milky Way, and Moon.

Cap all planet, satellite, comet, and asteroid names.

Cap Galaxy and Galactic IF author is referring to the Milky Way Galaxy (we need to distinguish the Milky Way from all other galaxies).

Cap Jovian, IF author is referring to the four planets (see below).

Planet names (possessive):

Earth, never say "the Earth". We don't say "the Jupiter" so Earth doesn't deserve any special privileges just 'cause we're here. Of course on Mars the martians probably say "the Mars" but we're better than them, right, so we won't be inconsistent.

Possessive planets: Try not to use the possessive form of Mars or Venus because they end with an awkward 's. If you must, then the proper form is one apostrophe with no "s": Venus' or Mars' or Uranus'.

Jovian (to cap or not to cap?): When referring to a specific characteristic of one planet use the modifier in lower case, such as martian surface or jovian atmosphere. There is one exception to this rule: The term "Jovian planets" is defined as the four giant gas planets (Jupiter, Saturn, Uranus, and Neptune). In this case, Jovian is capitalized.

Moon (or satellite) names:

Capitalize; possessive is one apostrophe if the name ends in "s," such as Iapetus' or Mimas'.

Meteorite names:

Martian meteorites: Originally these meteorites were called SNCs which stands for Shergottites, Nakhilites, and Chassignites after the first three meteorites that were grouped together and then classified as martian. However this acronym is rapidly losing favor because there have been many other meteorites, mostly from Antarctica, that are considered martian and they don't look like the SNCs so now folks prefer just calling them martian meteorites. So our standard will be "martian meteorites" unless there is a specific reason to use the term SNC.

Constellation names:

There has been a little bit of variations in the use of constellation names. There is a standard name, such as "Pegasus" and a genitive form, such as Pegasi. The genitive form is used when the author is referring to a specific star within the constellation, such as 51 Pegasi (a star in the constellation Pegasus). We should use the appropriate naming convention and then use the standard abbreviation throughout the text.

Geology

Epochs and Ages

- Initial cap names of formal historical epochs, geological ages, and strata. Formal terms are those included on the Geological Time Scale attached or their place equivalents (for example,

“Lower” equals “Early” in “Lower Jurassic” and thus would be capitalized) (for stratigraphy, we would use “Upper” and “Lower”; for time, we would use “Early” and “Late”).

Ex: Bronze Age, Stone Age, Lower Jurassic, Early Pliocene

- Lowercase informal terms.

Ex: Tertiary period, Cenozoic era

See the following references:

See PDF of pages from *Suggestions to Authors of the Reports of U.S. Geological Survey* (“Capitalization.USGS.Geological.Terms”).

Words into Type, p. 165.

Webster’s Third New International, p. 949, for geology chart.

But follow *Chicago*, 7.65 and 7.111, for *Science* style: Do not cap the words period, era, and epoch.

Soil Classification

- Cap names of particular soils, following *Words into Type* (p. 166) list, below:

Alpine Meadow	Laterite	Sierozem (Gray)
Bog	Pedalfer	Solonchak
Brown	Pedocal	Solonetz
Chernozem (Black)	Podzol	Soloth
Chestnut	Prairie	Terra Rossa
Desert	Ramann's Brown	Tundra
Gray-Brown Podzolic	Red	Wiesenboden
Half-Bog	Rendzina	Yellow

Water Types

- Cap the name of region but lowercase the term “aquifer.”

Ex: Floridian aquifer

Other Uses

Computer Terms

- Use of all caps or initial caps for programs and languages depends on (i) whether the word is an acronym or the name of a person or company and (ii) author preference.

Ex: Languages: BASIC, COBOL
Fortran (*Science* style)
Assembler, Pascal

Machinery: Cyber 205
Apple Macintosh
Fujitsu VP-200
Hitachi S-810
IBM PC ["personal computer" should be used unless
defined first time]

Commands: CANCEL, EXIT (use all caps)

See *Chicago* section 7.159 and 7.160.

Series

- Use of capitalization after a colon: The first word after a colon usually should be capped when it begins a complete sentence that is an important statement by itself.

Ex: The scenario leaves researchers in something of a bind: Either their long-term prediction is about to fail, or the prospects for short-term prediction will suffer a serious blow.

- In a series of items with a colon and (i), (ii), (iii), etc., cap the first word of each section if it is a complete sentence ending with a period; lowercase if not.

Ex: We assume the following: (i) that residues constitute part of the protein surface; (ii) that these residues are accessible to small reagents; and (iii) that the addition of a charged group to the residue would alter ion conduction.

The transcription factor E2F was considered a target for transcription activation for the following reasons: (i) E2F is a DNA binding protein. (ii) The duplicated E2F sites are critical for transcription. (iii) A single E2F site can confer regulation to a test promoter.

- Do not capitalize phrases or enumeration of items in series after a colon.

Ex: The following tools were used: hammer, wrench, and screwdriver.

Trademarks and Proprietary Names

- See the separate list of trademarks with reference book suggestions for checking trademarks. Do not use trademark symbols (®, ™) in any case. If possible, use the generic, rather than the trade, name. If a trade name must be used, capitalize it.

Ex: Dacron
Kleenex

LEGO uses all caps.

But: nylon, dry ice.

Models, Experiments, Equations

- Use lowercase for: model 1967
type 4
experiment 2
figure 1 in (34) [use only when referring to a figure in another report, article, etc.]
reaction 1

But: cap Eq. 2.

References

- Cap titles of books and journals (follow BGEA abbreviations), but cap only first word and proper nouns in titles of reports, which are in quotations.

Ex: “Assessment of carcinogenicity. . .,” *WHO Tech. Rep. Ser. 546* (1974).

- Lowercase “vol. 3,” “chap. 4,” “no. 7,” etc.
- Cap and abbreviate “editor” (Ed.).
- Lowercase and abbreviate “edition” (ed.).

See *Chicago* section 7.148 for treatment of motion pictures and TV or radio programs. Titles of paintings and sculpture are in section 7.154.

Supplementary Materials

- Spell out at all times in the text, lowercase.
- Cap names of online segments in the Supplementary Materials summary list after the references (e.g., Fig. S1, Table S1, Movie S1) but not as callouts in the text, which must be lower case (fig. S1, table S1, movie S1).